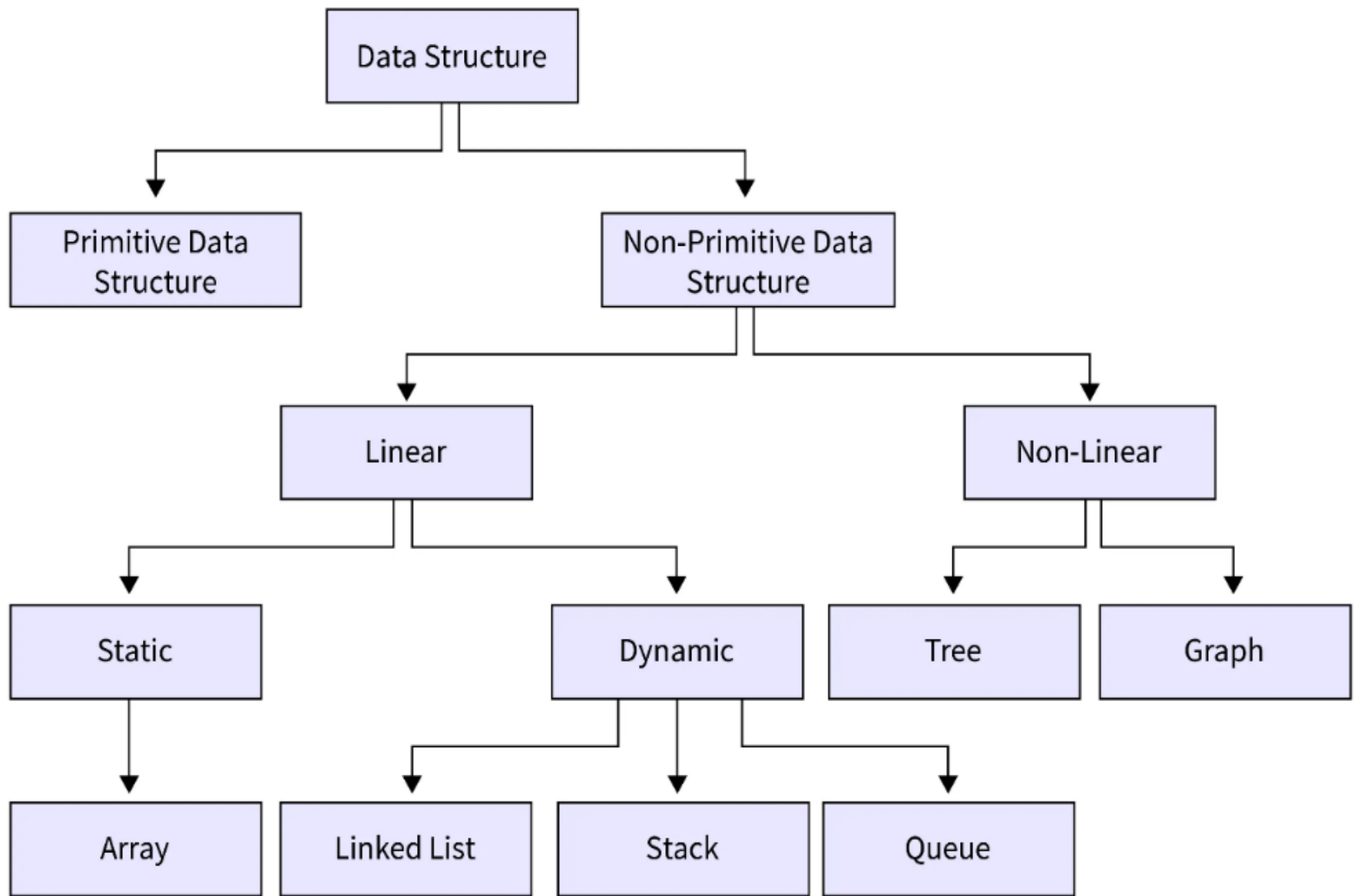
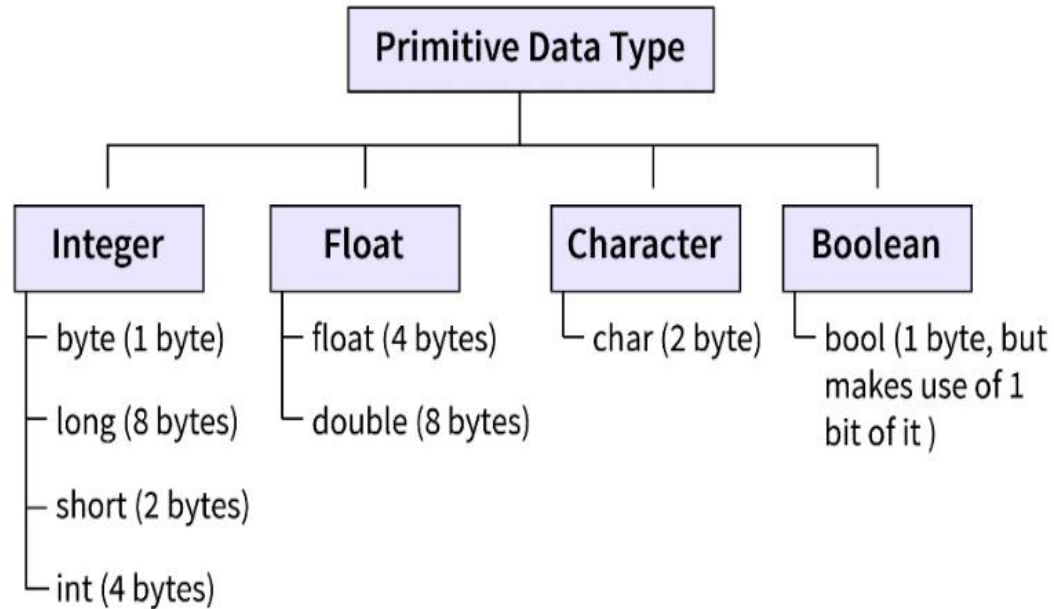




Remember – If Any data structure is implemented using Array then it will be always Static.



While calculating the Address in 1-D array or 2-D array, if the size of single element is not given and type of array is given then recall these sizes. (for example – in a numerical type of array is given as double then we must take size of single element = 8 bytes)

Size of Array

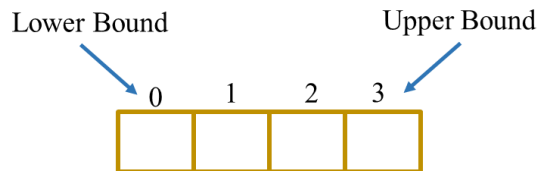
Number of Elements = Upper Bound – Lower Bound + 1

Upper Bound = Index of Last Element of Array

Lower Bound = Index of First Element of Array

The same
concept
for with
2D array.

Default Case

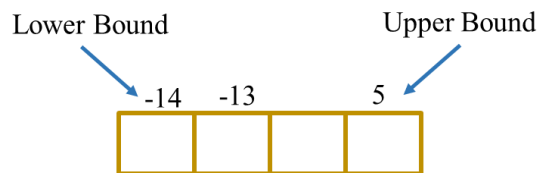


Number of Elements = Upper Bound – Lower Bound + 1

Number of Elements = $3 - 0 + 1 = 4$

Hence Total Elements = 4.

Take another Array A[-14....5]



Number of Elements = Upper Bound – Lower Bound + 1

Number of Elements = $5 - (-14) + 1$

Hence Total Elements = 20.

Address Calculation of One-Dimensional Array

$$\text{LOC}(A[i]) = \text{Base Address of } A + (i - \text{LB}) * \text{Size of Single Element}$$

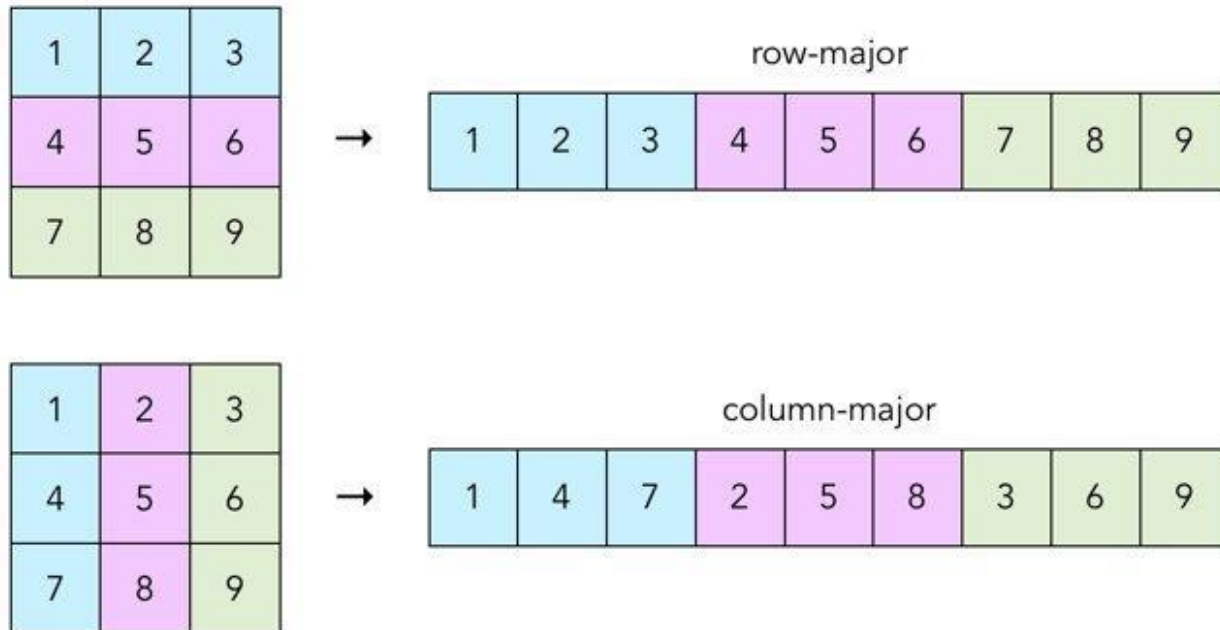
Where i is the Index at which you need to find the address and LB is the lower bound value of the Array.

Address Calculation of Two-Dimensional Array

In 2-D array, the data gets stored in 2 ways.

Row Major Order (RMO)

Column Major Order (CMO)



When Data Stored in RMO

$$\text{Address of } A[i][j] = B + W * ((i - LR) * N + (j - LC))$$

Where,

i = Row Subset of an element whose address to be found,

j = Column Subset of an element whose address to be found,

B = Base address,

W = Storage size of one element store in an array (in byte),

LR = Lower Limit of row/start row index of the matrix (If not given assume it as zero),

LC = Lower Limit of column/start column index of the matrix (If not given assume it as zero),

N = Number of columns given in the matrix.

What does the formula says

- (i-LR) = How many rows are there before ith element
- (j-LC) = How many columns are there before jth element
- [(i-LR)*N + (j-LC)] = How many element are there before i and j

When Data Stored in CMO

$$\text{Address of } A[i][j] = B + W * ((i - LR) + (j - LC) * M)$$

Where,

i = Row Subset of an element whose address to be found,

j = Column Subset of an element whose address to be found,

B = Base address,

W = Storage size of one element store in an array (in byte),

LR = Lower Limit of row/start row index of matrix(If not given assume it as zero),

LC = Lower Limit of column/start column index of matrix(If not given assume it as zero),

M = Number of rows given in the matrix.

What does the formula says

- $(i-LR)$ = How many rows are there before i^{th} element
- $(j-LC)$ = How many columns are there before j^{th} element
- $[(i-LR) + (j-LC)*M]$ = How many element are there before i and j

Address of k^{th} Element in 1-D Array

Let us take an example to understand this concept –

Consider an Array $A[-12.....25]$ and we want to find the address of the 19^{th} element.

In such situation use the below mentioned formula to find the index for the asked element.

$$\text{Index } [i] = L + K - 1$$

L is value of Lower bound (which is -12 in our case)

K is the element for which we are asked (which is 19 in our case)

Hence, index of 19^{th} element will be $= -12 + 19 - 1 = 6$

Now use the formula for 1 – D Address

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Address of k^{th} Element in 2-D Array

Let us take an example to understand this concept –

Consider an Array $P[-4...3][2...13]$ and we want to find the address of the 25th element.

In such situation,

First Check for CMO or RMO.

If CMO – then take L = Lower Bound of Row

If RMO – then take L = Lower Bound of Column

$$\text{Index } [i] = L + K - 1$$

Now use the formula for 1 – D Address